STUDY TITLE:

Empirical Evaluation of Creative Commons Understanding and Expectation

PRINCIPAL INVESTIGATOR:

Name: Aaron Shaw
Department: Communications Studies

STUDENT INVESTIGATOR (complete this section only if the project is student-initiated):

Name: Thomas Rousse
Department: Media, Technology and Society

Are you an:
☐ Undergraduate Student
☒ Graduate Student or Medical Student

VERSION DATE:

06/16/2020

RELATED STUDIES:

None.

Check any **applicable** boxes in the table below – you will be asked for further detail on these topics later in the protocol form:

Indicate Vulnerable Population(s) to be Enrolled	☐ Children (you must **complete Appendix A** in addition to this protocol document if you plan to enroll children) ☐ Cognitively Impaired Adults ☐ Pregnant Women (IF the research activities will affect the pregnancy or the fetus) ☐ Prisoners (or other detained/paroled individuals)
International Research (check this box if you will collect data from individuals located outside the United States)	(Methods will be used to screen by IP and identify spoofed addresses to remove international participants, but may not be 100% effective. No international participants will be intentionally included.) ☐
Research involving external collaborators (some research activities will be carried out by individuals not employed by Northwestern or NU affiliates)	☐
Research has U.S. Federal government funding via direct award or a sub-award (e.g., NIH, NSF, other federal agencies or departments)	☐

1.0 Purpose and rationale of the study:

Creative Commons is an open-licensing project created more than twenty years ago that aims to give content creators more granular control over their ownership rights associated with copyright.

Using a between-subject vignette survey design, this study will compare materials marked with normal copyright protections versus a selection of Creative Commons licenses. By comparing results between the control group (with standard copyright protection) and Creative Commons, this study will provide causal empirical evidence about whether Creative Commons increases the likelihood of infringement and whether lay users of

copyright are responsive to civil and criminal penalties associated with copyright protection. Basic demographic information will also be collected.

2.0 Enrollment Criteria (who can be in your study and who would not be eligible to participate in your study):

This study assumes that the potential population of American Creative Commons licensors is functionally equivalent to the U.S. adult population as a whole. A sample of the adult U.S. population will be collected through the crowd-sourcing platform Mechanical Turk ("MTurk") and then weighted to approximate a true, randomized national sample. As others have noted, the potential to commit copyright infringement, or to have created a work that is easily available for infringement, is far less rarefied than it was before the advent of widely available digital technology connected to the Internet. Furthermore, given that the only prerequisite for copyright is that the form of expression is fixed, virtually every U.S. citizen will create a copyrighted work during their lifetime and thus could avail themselves of a Creative Commons license. Concerns that a general audience is unlikely to accurately grasp a topic with the depth and complexity of copyright are likely well-founded. However, copyright follows the doctrine of strict liability: ignorance or good intentions are not a defense, although willful behavior is punished more harshly.

No vulnerable populations will be targeted for this study.

3.0 Sample Size:

For this survey, roughly 1,000 participants will be recruited, with allowances for collecting additional survey samples for incomplete or illegitimate responses. The sample will be restricted to American adults (age 18+) who can read and write English. Participants will be drawn from Amazon's Mechanical Turk service with additional criteria and checks to prevent participants from other countries from concealing their location in order to participate, discussed below. This sample size will provide ample statistical power.

4.0 Recruitment and Screening Methods:

A sample of the adult U.S. population will be collected through the crowd-sourcing platform Mechanical Turk ("MTurk") and then weighted to

approximate a true, randomized national sample. Crowdworkers on MTurk are given discrete tasks, called Human Intelligence Tasks ("HITs") in exchange for compensation; social scientists have used MTurk as an inexpensive, fast survey recruitment method for more than a decade.[1] To lessen the impact of the time of data collection on survey population composition and to reduce costs, the survey will use the "microbatching" feature of a third-party management system that spreads out data collection over time.[2] The survey itself will be administered using Qualtrics, subscription-based software for collecting survey results for academic projects, market research, and other tasks. MTurk workers are provided with a link to the Qualtrics survey after signing up for the HIT; upon successful completion of the survey, Qualtrics provides a unique code to enter on MTurk to ensure that workers receive credit. Qualtrics provides affordances for randomizing assignment between control and treatment groups, question order, recording response times, and more.

Recently, MTurk has suffered from a "quality crisis" caused by automated responses (i.e. "bots") and foreign respondents incorrectly identifying themselves as Americans and circumventing technological controls.[3] As a result, it is necessary to take additional precautions to maintain data quality. A team of researchers has developed a protocol for Qualtrics that integrates looking up respondent IP against a "blacklist" of known IP addresses that individuals in other countries use to appear to respond from the U.S.[4] Standard quality control methods in MTurk survey research also include limiting participants to workers Amazon has certified as "masters" and requiring 95% or higher successful completion of HITs. Respondents will also be required to successfully answer "attention check" questions that deter random responses.

---

[1] *See generally* Kim Bartel Sheehan, *Crowdsourcing Research: Data Collection with Amazon's Mechanical Turk*, 85 COMMUNICATION MONOGRAPHS 140 (2017).

[2] Leib Litman, Jonathan Robinson, & Tzvi Abberbock, *TurkPrime.com: A Versatile Crowdsourcing Data Acquisition Platform for the Behavioral Sciences*, 49 BEHAVIOR RESEARCH METHODS 433 (2017) (describing efforts by the authors to develop the TurkPrime platform). *See also* Steven Komarov, Katharina Reinecke, & Krzysztof Z. Gajos, *Crowdsourcing Performance Evaluations of User Interfaces*, 2013 PROCEEDINGS OF THE SIGCHI CONFERENCE ON HUMAN FACTORS IN COMPUTING SYSTEMS 207, 214 (2013) (finding the time of day or part of week had significant effects on results).

[3] *See, e.g.*, Emily Dreyfuss, "A Bot Panic Hits Amazon's Mechanical Turk," WIRED, (Aug. 17 2018, 11:38 AM) (explaining early detection and attempts to explain the "bot scare") https://www.wired.com/story/amazon-mechanical-turk-bot-panic/ (last visited Feb. 9, 2019).

[4] Ryan Kennedy, Scott Clifford, Tyler Burleigh, Ryan Jewell, & Philip Waggoner, *The Shape of and Solutions to the MTurk Quality Crisis*, (Oct. 24, 2018) (advising the use of IP scanning software IP Hub and providing instructions on how to implement it within Qualtrics) https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3272468

5.0 Research Locations:

Participants will use their own computer equipment at a location of their choice to access the Mechanical Turk website, where they will be linked to consent forms and the survey on Qualtrics. Participants will be screened by IP address blacklists for known VPNs and proxies used to defeat MTurk's restrictions on participation.

6.0 Multi-site Research (research that involves external collaborating institutions and individuals):

Not applicable.

7.0 International Research (where data collection will occur outside the United States and U.S. territories, including online activities)

Not applicable. All participants will be screened for US-originating IP addresses, along with a blacklist of IP addresses commonly used to spoof a US point of origin.

8.0 Procedures Involved:

Please check the boxes for all applicable data collection procedures you plan to use:

☐One-on-one interviews
☐Focus Groups
☒Questionnaires/surveys
☐Analysis of secondary data (medical record data, educational records, government or private sector datasets, etc.)
☐Ethnographic observation
☐Physiological measurements (e.g., EEG, EKG, MRI)
☐Biospecimen collection (saliva samples, blood draws, hair samples, etc.)
☐Mobile applications/data collection devices (e.g., Fitbits, actigraphs, etc.)
☐Behavioral decisionmaking tasks (e.g., puzzles, interactive games, etc.)
☐Physical activities such as walking and other forms of exercise
☐Other procedures (briefly list types of procedures here if not covered by the check-boxes above): _____

Participants will complete the survey using their own computer equipment in a location of their choice. Participants will be able to complete

the survey at any day and time that they desire while the survey is made available in small batches. Participants will access the survey via a link provided in the study advertisement on MTurk. The survey will be hosted by Qualtrics, and participants will complete the entire survey on that platform.

Consent will be obtained prior to participants beginning the survey. In lieu of a signature, participants will type a short phrase. [See attached Consent Form].

All participants will complete basic demographic information including age, sex, ethnicity, and educational attainment. Data will be assigned a random ID number generated by Qualtrics, and no further identifying information will be collected or retained, rendering responses to the survey anonymous.

For the vignette portion of the study, participants will answer a series of questions about whether they consider use of creative material copyright infringement, likely consequences, and their intuition regarding appropriate punishment. [See attached Questionnaire]. Answering all questions in the survey will take a mean time of 15 minutes.

At the conclusion of the survey, participants will be provided with an educational debriefing script [see end of attached Questionnaire], including a reminder of the contact information for the IRB and primary investigator in the event of questions or a desire to follow-up after the study.

This study poses minimal risk to no risk to participants. No question asks participants to report their involvement in sensitive activities, nor is there any reason to believe that participants will find the survey to be stressful. The vignettes are hypothetical examples and no copyright infringement will be committed by participants.

The project is expected to last roughly six weeks, and the majority of data collection will likely occur over just a few days given high availability of MTurk workers. Since this is a survey-based study, any individual participant is going to be involved only for however long it takes them to complete the questionnaire. This is estimated to be an average of fifteen minutes.

9.0 Research with Vulnerable Populations (if children are the ONLY vulnerable population you plan to enroll, do NOT complete this section – instead fill out Appendix A)

No vulnerable population will be intentionally included in this study, and in the event any respondents are inadvertently identified as a member of a vulnerable population, their responses and any collected information will be destroyed.

10.0 Incomplete Disclosure or Deception:

This study will not use incomplete disclosure or deception.

11.0 Consent Process:

Consent will be obtained via a consent form programmed into the preliminary stages of the online survey. Participants will have unlimited time to review the consent script, will be provided with contact information for the primary investigator and the IRB in the case of questions about the consent process and/or the study in general. Participants will have the option to decline consent on the online form and immediately exit from the survey. If participants agree to participate, they will be encouraged to print or otherwise save a copy of the consent script before they begin the survey questions. The consent process should take no more than 5 minutes to complete. All participants will be adult (18+ years) English speakers, so we anticipate no problems with participants being able to read and comprehend the nature of consent being granted for this survey. No participants from impaired or limited capacity populations are being targeted for recruitment. Further, the option to decline to consent and leave the survey or agree to participate and begin the survey are clearly spelled out in the script and require an affirmative response by the participants to select the option that best fits their response. [See attached Consent Form].

Participants' consent will be recorded one way or the other (agree or decline) in the data compiled by Qualtrics, the online platform hosting the survey. No other data will be collected if a participant declines to participate. The consent script will be presented to participants, and it is on this page of the programmed online study that they will also indicate their consent (or lack thereof). Participants will be encouraged to print or save a copy of the consent script before making their decision. [See attached Consent Form].

12.0 Waiver of Participant Signature on Consent Form:

Because the primary method of administration is an online survey, participants will indicate their consent via a digital acknowledgement and typing their name through the Qualtrics survey interface.

13.0 Waivers and Alterations of Consent Information:

No waiver or alternation of consent is requested beyond the waiver of a physical signature discussed above.

14.0 Financial Compensation:

While part of the appeal of crowd-sourced survey results are their low cost, several researchers and commentators have expressed ethical concerns about the very low payment rates common on crowdsourcing platforms like Mechanical Turk.[5] By one estimate, a quarter of crowd workers make more than three quarters of their income through crowdsourcing platforms, but Amazon and other providers classify participants as independent contractors, not employees, and are therefore exempt from paying minimum wage.[6] To partially address this ethical concern, respondent pay will be adjusted to roughly 110% of the current U.S. federal minimum wage, or $8 per hour. Because this is significantly more than the prevailing wage on MTurk,[7] and higher compensation may attract an unpredictable subset of the community or disproportionately incentivize deceptive behavior,[8] the survey will be advertised and initially compensated at the prevailing rate on MTurk, and after data collection is complete, respondent pay will be adjusted through bonuses so total compensation is based on the target pay-rate by time spent on the survey.[9] Amazon charges a base rate of 20% of any reward or bonus,

---

[5] *See, e.g.*, Vanessa Williamson, *On the Ethics of Crowdsourced Research*, 49 PS: POLITICAL SCI. & POLITICS 77 (2016).

[6] M.S. Silberman et al., *Responsible Research with Crowds: Pay Crowdworkers at Least Minimum Wage*, 61 COMMUNICATIONS OF THE ASSOC. FOR COMPUTING MACHINERY 39 (2018).

[7] Average compensation in a recent study was $3.13 per hour. Only 4% of workers earned more than minimum wage. Kotaro Haro et al., *A Data-Driven Analysis of Workers' Earnings on Amazon Mechanical Turk*, Computer Human Interaction (CHI) Conference, (Apr. 21–26, 2018).

[8] For example, participants who do not meet the minimum prerequisites regarding age or location could use incorrect profile information or technological controls to qualify. Alternatively, a single user could operate multiple accounts to receive compensation more than once.

[9] *See* Williamson, *supra* note 5 (describing the use of automation to facilitate bonus payments for a subject pool and linking to example scripts written for this purpose). Waiting until data collection is complete removes the risk that MTurk workers will communicate the actual compensation rate to other workers. *See* Gordon B. Schmidt, *Fifty Days an MTurk Worker: The Social and Motivational Context for Amazon Mechanical Turk Workers*, 8 INDUSTRIAL AND ORGANIZATION PSYCHOLOGY 165 (2015) (describing participant observation of MTurk communities and information-sharing among crowdworkers). Concealing

with an additional 5% charge for selecting "Master" respondents who have a track record of acceptable task completion, and an additional 20% for each batch of questions seeking ten or more respondents.[10] Assuming the use of TurkPrime's microbatching feature and Master respondents, Amazon's fee will be 25%. A previous research survey featuring nine vignettes of 25–50 words, demographic questions, and a short survey measure with roughly 1,000 respondents cost about $5,000.[11] Assuming a fifteen-minute mean survey completion time, collecting 1,000 responses through MTurk will cost about $2,500.[12]

15.0 Audio/Video Recording/Photography

No audio/video recording or photography will be taken as part of this research.

16.0 Potential Benefits of this Research:

There are no direct benefits to participants for completing the survey beyond learning indirectly about Creative Commons and copyright protection available to them.

The results of this survey will provide empirical answers to whether Creative Commons licensing encourages infringement, and may provide Creative Commons advocates and others with information to improve open licensing of copyrighted materials.

17.0 Potential Risks to Participants:

Participants will be subject to minimal risk for this study. There is a small chance that information they provide will be inadvertently disclosed by researchers, but no identifying information will be collected. Compensation may be delayed or not delivered due to technical difficulties or failure to follow instructions. Bad faith participation or failure to comply with instructions may result in expulsion from the survey, and a resulting

---

a time-based pay rate also prevents deliberately long completion times for greater compensation (i.e. "goldbricking").

[10] *See* "Amazon Mechanical Turk Pricing," Amazon MTurk https://requester.mturk.com/pricing last accessed Feb. 15, 2019.

[11] Matthew B. Kugler & Thomas H. Rousse, *The Privacy Hierarchy & Trade Secret Expectations*, 104 IOWA L. REV. 1223 (2019).

[12] Respondents (1,000) x Time (.25) x hourly rate (8) x Amazon fee (1.25) = $2,500.

loss of income or opportunities to participate in future studies. If participants do experience discomfort or for any reason want to decline to answer specific questions and/or complete the survey, they will be free to do so with no consequences for them. During the consent process, participants will learn that they may leave the survey at any time. If a participant terminates participation after beginning the survey, no additional data will be collected, and any data already collected will be deleted and will not be used for data analyses.

18.0 Provisions to Protect Participant Privacy and Data Confidentiality:

Qualtrics is a secure platform that encrypts data transmission between the service and respondents. No participant identifiers will be collected or stored for respondents. MTurk user names that could be used to associate respondents with other responses will be removed from datasets prior to analysis and long-term storage and replaced with numerical codes. Data will be downloaded from Qualtrics and stored on a password-protected server as well as physical media secured by the student investigator in a private office. All responses will be subsequently deleted from Qualtrics. Data will be kept for at least three years, and may be stored by a third party in the future as required by publishing agreements for the validation and reproduction of research.

19.0 Data Monitoring Plan to Ensure the Safety of Participants:

The data collected in this study does not pose a threat to the safety of participants.

20.0 Long-term Data and Specimen Storage and Sharing:

Anonymized data will be stored and shared for long-term future research studies or replication as required by any future publication venue.

No identifiers will be included with the data when shared.

21.0 Qualifications of Research Team to Conduct the Research:

The primary investigator is an assistant professor in the School of Communication with an extensive publication record. The lead

student researcher is a J.D./Ph.D. Candidate with training in qualitative methods and who has conducted several surveys involving vignettes, MTurk, and/or Qualtrics, including one recently published law review article.[13]

[13] Matthew H. Kugler & Thomas H. Rousse, *The Privacy Hierarchy: Trade Secret and Fourth Amendment Expectations*, 104 IOWA LAW REVIEW 1223 (2019).